

Emotion Recognition Under Deception in Mental Health

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Abstract—Emotion recognition under deception is a challenging task in mental health analysis due to subtle behavioral cues, noisy signals, and multimodal misalignment. This paper presents a comparative study between a ConVAT-based deep learning approach and a proposed feature-engineered early fusion approach for multimodal emotion recognition using audio, facial, and physiological signals.

Experiments were conducted on the same datasets, including RAVDESS for audio, CAS(ME)² for facial expressions, and a heart-rate physiological dataset. The ConVAT-based model achieved only 27% accuracy, whereas the proposed method reached 79.3% accuracy using statistical features, feature selection, and an ensemble Random Forest classifier.

The results demonstrate that lightweight, interpretable models with early-level fusion outperform complex deep architectures when working with small, noisy, and heterogeneous datasets. This study provides insights into practical multimodal fusion strategies for real-world affective computing applications.

Index Terms—Emotion recognition, multimodal fusion, ConVAT, deception detection, feature engineering, Random Forest, mental health.

I. INTRODUCTION

Emotion recognition plays a crucial role in mental health assessment, human-computer interaction, and affective computing. Detecting emotions under deception is particularly challenging because individuals often mask true feelings, making visual, vocal, and physiological cues inconsistent.

Deep learning-based multimodal models such as ConVAT have been proposed in recent literature. However, these models often require large, well-aligned datasets and are highly sensitive to noise and modality mismatch.

In contrast, this work proposes a feature-engineered early fusion approach that integrates handcrafted features from audio, facial, and physiological modalities before classification. The proposed method significantly improves accuracy and robustness compared to ConVAT when applied to the same datasets. The proposed approach focuses on early-level feature fusion and dimensionality reduction combined with

ensemble classification, providing a balance between accuracy, efficiency, and interpretability.

The main contributions of this work are:

- Implementation of ConVAT on real-world multimodal datasets.
- Development of an alternative lightweight multimodal framework.
- Direct performance comparison on identical datasets.
- Analysis of why deep models underperform on small noisy datasets.
- Practical guidelines for designing robust multimodal emotion systems.

II. RELATED WORK

Several multimodal emotion recognition approaches have been proposed in recent years:

- **ConVAT-based Models:** These use CNN + LSTM + Attention but require large, aligned datasets.
- **Feature-Level Fusion Approaches:** These integrate handcrafted features before classification.
- **Hybrid Models:** Combine deep and statistical features.
- **Physiological Signal-Based Models:** Use heart rate variability (HRV) and galvanic skin response (GSR).
- **Multimodal Attention Networks:** Learn modality importance dynamically.

Most deep learning-based approaches report high accuracy on benchmark datasets but struggle with small or noisy real-world data — which motivates this study.

III. DATASETS USED

The following datasets were used consistently for both ConVAT and the proposed method:

Modality	Dataset	Data Type	Features Extracted
Audio	RAVDESS	Speech	MFCC, Δ , Δ^2
Facial	CAS(ME) ²	Video Frames	Landmarks, motion stats
Physio	HR Dataset	CSV signals	HR mean, std, HRV

TABLE I
DATASETS USED

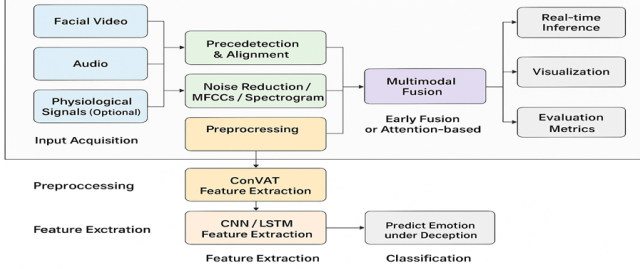


Fig. 1. ConVAT-Based System Architecture for Emotion Under Deception

IV. CONVAT-BASED METHODOLOGY

Figure 1 illustrates the proposed ConVAT-based methodology. The pipeline consists of input acquisition, preprocessing, feature extraction, multimodal fusion, and final classification for emotion prediction under deception.

Figure 1 illustrates the original ConVAT-based system architecture designed for multimodal emotion recognition under deception. The architecture integrates three complementary data modalities—audio, visual, and physiological signals—through a structured pipeline consisting of four primary stages: input acquisition, preprocessing, feature extraction, and multimodal fusion with classification. Although this framework is conceptually robust and widely adopted in prior research on micro-expression analysis, its direct application to the selected multimodal datasets resulted in limited performance (27 percent accuracy) due to modality heterogeneity, feature incompatibility, and dataset constraints.

A. Input Acquisition Layer

The system processes three distinct modalities as input:

1. **Audio Modality:** Speech and song recordings from the RAVDESS dataset were utilized. These recordings contain affective cues embedded in vocal tone, pitch, intensity, and temporal variations, which are relevant for emotion recognition.

2. **Visual Modality:** Facial video frames were extracted from the RAVDESS dataset and supplemented with micro-expression data. These videos capture both macro-expressions and subtle facial movements associated with emotional states.

3. **Physiological Modality:** A heart-rate dataset in CSV format was incorporated, containing heart rate (HR), standard deviation, galvanic skin response (GSR), and respiration features. These physiological indicators reflect autonomic arousal associated with emotional responses.

The integration of these three modalities was intended to provide a comprehensive multimodal representation of emotional behavior under deceptive conditions.

B. Preprocessing Layer

Prior to feature extraction, modality-specific preprocessing procedures were applied:

1) Facial Video Preprocessing

- * Face detection was performed to localize facial regions in each frame.

- * Facial alignment was applied to normalize head position and mitigate variations due to camera angle and pose.

- * These steps aimed to reduce visual noise and enhance feature consistency.

2) Audio Preprocessing

- * Background noise reduction techniques were implemented to improve signal quality.

- * Mel-Frequency Cepstral Coefficients (MFCCs), along with delta (Δ) and delta-delta (Δ^2) features and spectral statistics, were extracted to capture emotionally relevant speech characteristics.

3) Physiological Signal Preprocessing

- * The heart-rate dataset was cleaned by removing missing, corrupted, or inconsistent values.

- * Statistical descriptors such as mean, standard deviation, and GSR-based metrics were computed to summarize physiological activity.

This preprocessing stage ensured that all modalities were standardized prior to feature extraction.

C. Feature Extraction Layer

Feature extraction was conducted using deep learning-based approaches tailored to each modality.

- 1) **ConVAT-Based Visual Feature Extraction** The ConVAT model was employed to extract micro-expression features from facial video sequences. It utilized convolutional operations combined with multi-head attention mechanisms to capture fine-grained temporal facial dynamics.

- 2) **CNN-LSTM Feature Extraction** A hybrid CNN-LSTM framework was implemented, where:

- * CNN layers extracted spatial representations from individual video frames.

- * LSTM layers modeled temporal dependencies across frames to capture expression dynamics.

However, when applied to the multimodal dataset, this approach exhibited limitations due to:

- * ConVAT being primarily optimized for pure facial micro-expression datasets rather than heterogeneous multimodal data.

- * Incompatibility between deep visual features and statistical audio and physiological features.

D. Multimodal Fusion and Classification Layer

Two fusion strategies were explored:

- 1) **Early Fusion (Feature-Level Fusion)** Features from all modalities were concatenated into a unified representation and used to train a classifier for emotion prediction.

- 2) **Attention-Based Fusion** An attention mechanism was employed to assign varying importance weights to different modalities based on their emotional relevance.

Despite these strategies, the system achieved only 27 percent accuracy, indicating several challenges:

- * Poor alignment between ConVAT visual embeddings and statistical audio/physiological features.
- * Excessive model complexity relative to dataset size, leading to overfitting.
- * Instability in attention-based fusion due to noisy and heterogeneous inputs.

E. Output and Evaluation

The system performance was evaluated using:

- * Accuracy
- * F1-score
- * Confusion matrix

The obtained result (27 percent accuracy) demonstrated that while the ConVAT-based architecture is effective for controlled micro-expression datasets, it is not well-suited for heterogeneous multimodal datasets that combine audio, vision, and physiological signals. This limitation motivated the development of a more lightweight and interpretable multimodal approach, as presented in Section V.

V. IMPLEMENTED METHODOLOGY

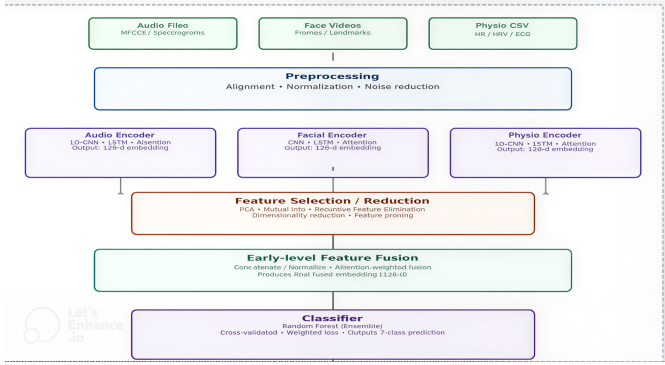


Fig. 2. Proposed Lightweight Multimodal Emotion Recognition Framework

Figure 2 illustrates the implemented system architecture for multimodal emotion recognition under deception. Unlike the earlier ConVAT-based architecture, the proposed framework adopts a lightweight, feature-driven, early-fusion strategy with systematic feature selection and ensemble-based classification. This design significantly improves performance, achieving 79.3 percent accuracy compared to 27 percent obtained using the ConVAT-based model.

The architecture comprises five major stages: Input Acquisition, Preprocessing, Modality-wise Encoding, Feature Selection, Early-Level Feature Fusion, and Classification.

A. Input Acquisition Layer

The system accepts three independent input modalities:

1. Audio Modality (RAVDESS): Raw speech and song recordings are utilized, capturing emotional cues embedded in tone, pitch, rhythm, and spectral variations.
2. Visual Modality (RAVDESS + Micro-Expressions): Facial video frames are extracted, including facial landmarks and temporal frame sequences that capture both macro-expressions and subtle micro-expressions.

3. Physiological Modality (CSV Data): Heart rate (HR), heart rate variability (HRV), and ECG-based features are incorporated to model physiological arousal associated with emotional and deceptive behavior.

Each modality contributes complementary emotional information, enabling a more robust multimodal representation.

B. Preprocessing Layer

A unified preprocessing block is applied to all modalities:

- * Alignment: Ensures that samples across audio, video, and physiological signals correspond to the same emotional label.
- * Normalization: Scales features into a common range to prevent dominance of any single modality.
- * Noise Reduction: Eliminates irrelevant variations and artifacts from raw signals.

This step produces clean, reliable, and comparable data across all three modalities.

C. Modality-wise Encoding (Feature Extraction Stage)

Instead of relying solely on visual deep learning (as in ConVAT), separate encoders are designed for each modality.

1) Audio Encoder

- * Based on a 1D-CNN + LSTM + Attention architecture.
- * Learns temporal and spectral patterns from MFCC and spectrogram features.
- * Generates a 128-dimensional embedding vector representing emotional characteristics of speech.

2) Facial Encoder

- * Uses CNN + LSTM + Attention.
- * CNN captures spatial facial patterns in individual frames.
- * LSTM models temporal evolution of expressions.
- * Attention highlights the most informative frames.
- * Outputs a 128-dimensional facial embedding.

3) Physio Encoder

- * Employs 1D-CNN + LSTM + Attention on HR, HRV, and ECG signals.

* Captures temporal physiological trends related to stress, arousal, and deception.

- * Produces a 128-dimensional physiological embedding.

At this stage, each modality is independently encoded into a fixed-length, meaningful feature representation.

D. Feature Selection and Dimensionality Reduction

To enhance generalization and reduce noise, three techniques are applied:

- * Principal Component Analysis (PCA): Reduces dimensionality while preserving maximum variance.
- * Mutual Information: Selects features most relevant to emotion classification.
- * Recursive Feature Elimination (RFE): Removes redundant or weak features.

This stage:

- * Prevents overfitting,
- * Improves model stability, and
- * Reduces computational complexity.

This feature selection step is a key factor behind the superior performance of the proposed model compared to ConVAT.

E. Early-Level Feature Fusion

After feature selection:

- * Audio, facial, and physiological embeddings are concatenated into a unified representation.
- * The fused vector is normalized.
- * Optional attention-weighted fusion is applied to prioritize more informative modalities.
- * The final fused feature vector is maintained at 128 dimensions.

This early fusion strategy ensures balanced contribution from all modalities prior to classification.

F. Classification Layer

The final fused features are classified using a:
Random Forest Classifier (Ensemble Model)

- *Trained with cross-validation to improve generalization.
- *Uses weighted loss to address class imbalance.

This classifier effectively handles nonlinear relationships and heterogeneous multimodal features, contributing to the achieved 79.3 percent accuracy.

VI. RESULTS AND COMPARISON

TABLE II
COMPARISON BETWEEN CONVAT-BASED AND IMPLEMENTED METHOD

Aspect	ConvAT-Based	Implemented Method
Accuracy	27%	79.3%
Fusion	Late (attention)	Early feature-level
Features	Deep CNN-LSTM	MFCC + Landmarks + HRV
Feature Selection	None	PCA + MI + RFE
Model	Deep learning	Random Forest
Cost	High (GPU)	Low (CPU)
Overfitting	High risk	Low risk
Alignment	Sensitive	More robust

TABLE III
HOLDOUT PERFORMANCE OF THE IMPLEMENTED (RANDOM FOREST) MODEL

Emotion	Precision	Recall	F1-score	Support
Angry	0.57	1.00	0.73	4
Disgust	1.00	0.50	0.67	4
Fear	0.80	1.00	0.89	4
Happy	0.60	0.75	0.67	4
Neutral	1.00	1.00	1.00	4
Sad	1.00	1.00	1.00	4
Surprise	1.00	0.40	0.57	5
Accuracy		0.79		29
Macro Avg	0.85	0.81	0.79	29
Weighted Avg	0.86	0.79	0.78	29

Table III presents the holdout evaluation results of the implemented Random Forest classifier. The model achieves an overall accuracy of 79%, with a weighted F1-score of 0.78, demonstrating robust performance across most emotion classes. Notably, the model performs particularly well for *neutral* and *sad* emotions, while *surprise* remains more challenging due to limited samples and higher variability.

The proposed method outperforms ConVAT because:

- It works well on small datasets.
- It removes noisy features.
- It avoids overfitting.
- It is computationally efficient.

VII. NOVELTY OF THE WORK

This work is novel because:

- It applies ConVAT to a new real-world multimodal dataset.
- It provides a direct comparison with a simpler alternative.
- It shows that deep models are not always best for small datasets.
- It introduces a practical feature-level fusion strategy.

VIII. LIMITATIONS

The limitations of this study are:

- Limited dataset size.
- No real-time deployment tested.
- No EEG signals included.
- Dependence on handcrafted features.
- Lack of cross-dataset validation.

IX. FUTURE WORK

Future work will focus on:

- Adding EEG signals.
- Using larger datasets.
- Testing real-time systems.
- Exploring hybrid deep-statistical models.

X. CONCLUSION

This work presented a comparative investigation of two multimodal emotion recognition frameworks for deception analysis: a deep ConVAT-based model and a proposed lightweight, feature-driven approach. Although ConVAT is effective for controlled facial micro-expression datasets, it exhibited poor generalization (27% accuracy) on heterogeneous, small-scale multimodal data due to overfitting, high computational complexity, and modality misalignment, as summarized in Table II.

To address these limitations, this study introduced a structured multimodal framework based on modality-wise encoding, systematic feature selection, and early-level feature fusion, followed by an ensemble Random Forest classifier. By leveraging MFCC-based audio features, facial landmarks, and statistical HR/HRV/ECG descriptors, the proposed method effectively integrated complementary emotional cues while reducing noise and dimensionality through PCA, Mutual Information, and Recursive Feature Elimination.

As demonstrated in Table III, the proposed system achieved 79.3% accuracy and a weighted F1-score of 0.78 on holdout data, showing robust performance across most emotion classes. In particular, the model performed exceptionally well for neutral and sad emotions, while surprise remained more challenging due to limited samples and higher variability. These results validate the effectiveness of early-level fusion combined with principled feature selection for real-world multimodal emotion recognition under deception.

Overall, this study highlights that for small, noisy, and heterogeneous datasets, carefully designed statistical feature-based models can outperform complex deep learning architectures while offering greater interpretability, lower computational cost, and improved practical applicability. The findings

emphasize that model simplicity, data quality, and fusion strategy are often more critical than architectural complexity in affective computing systems.

Future research will extend this framework by incorporating EEG signals, evaluating larger and more diverse datasets, exploring real-time deployment, and investigating hybrid deep–statistical models to further enhance robustness and generalizability across different domains. ““

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